

Instructions: Use proper notation, and show all of your work to receive credit. Round numeric answers to at least 2 decimal places.

1. (2 pts) Which of the following is not a property of the cdf, $F(y)$? Handout 12, pg. 1

(a) $\lim_{y \rightarrow -\infty} F(y) = 0$.

(b) $\lim_{y \rightarrow \infty} F(y) = 1$.

(c) $F(y)$ is nondecreasing.

(d) $F(y)$ is left-continuous.

2. (2 pts) True or False (circle your answer): The chi-square and exponential distributions are two common types of the gamma distribution. Handout 15, pp. 1-2

True

False

3. (3 pts) Suppose that Y has density function

$$f(y) = \begin{cases} ky(1-y), & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Section 4.1-4.2,
Problem 8a

Find the value of k that makes $f(y)$ a probability density function.

$$\begin{aligned} \int_0^1 ky(1-y) dy &= 1 \Rightarrow k \int_0^1 (y - y^2) dy = 1 \\ &\Rightarrow k \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 = 1 \\ &\Rightarrow k \left[\frac{1}{2} - \frac{1}{3} \right] = 1 \\ &\Rightarrow k \left(\frac{1}{6} \right) = 1 \\ &\Rightarrow \boxed{k = 6} \end{aligned}$$

4. (3 pts) If Z is a standard normal random variable, find the value of z_0 such that $P(-z_0 < Z < z_0) = .99$. Section 4.5, Problem 59d

$$z_0 = \text{invNorm}(.995, 0, 1) \approx 2.576$$

5. (5 pts) A supplier of kerosene has a 150-gallon tank that is filled at the beginning of each week. His weekly demand shows a relative frequency behavior that increases steadily up to 100 gallons and then levels off between 100 and 150 gallons. If Y denotes weekly demand in hundreds of gallons, the relative frequency of demand can be modeled by

$$f(y) = \begin{cases} y, & 0 \leq y \leq 1 \\ 1, & 1 < y \leq 1.5 \\ 0, & \text{elsewhere} \end{cases}$$

Find $F(y)$. Section 4.1-4.2, Problem 13a

Case 1 ($y < 0$): $F(y) = P(Y \leq y) = \int_{-\infty}^y 0 dt = 0$

Case 2 ($0 \leq y \leq 1$): $F(y) = P(Y \leq y) = \int_{-\infty}^0 0 dt + \int_0^y t dt = \frac{1}{2} t^2 \Big|_0^y = \frac{1}{2} y^2$

Case 3 ($1 < y \leq 1.5$): $F(y) = P(Y \leq y) = \int_{-\infty}^0 0 dt + \int_0^1 t dt + \int_1^y 1 dt$
 $= \frac{1}{2} t^2 \Big|_0^1 + t \Big|_1^y$
 $= \frac{1}{2} + (y - 1)$
 $= y - \frac{1}{2}$

Case 4 ($y > 1.5$): $F(y) = P(Y \leq y) = \int_{-\infty}^0 0 dt + \int_0^1 t dt + \int_1^{1.5} 1 dt + \int_{1.5}^y 0 dt$
 $= \frac{1}{2} + (1.5 - 1)$
 $= 1$

$$F(y) = \begin{cases} 0, & y < 0 \\ \frac{1}{2} y^2, & 0 \leq y \leq 1 \\ y - \frac{1}{2}, & 1 < y \leq 1.5 \\ 1, & y > 1.5 \end{cases}$$

6. (3 pts) Suppose $Y \sim \exp(10)$. Find $P(Y < 3)$. You may use the fact that the cdf of Y is

$$F(y) = \begin{cases} 0, & y < 0 \\ 1 - e^{-y/10}, & y \geq 0 \end{cases}$$

} Handout 15,
pg. 3

$$P(Y < 3) = F(3) = 1 - e^{-3/10} \approx .2592$$

7. (4 pts) Suppose $X \sim U(a, b)$. Prove that $E(X) = \frac{a+b}{2}$. Handout 13, pg. 2

$$\begin{aligned} E(X) &= \int_a^b x \cdot \frac{1}{b-a} dx \\ &= \frac{1}{b-a} \left[\frac{1}{2} x^2 \Big|_a^b \right] \\ &= \frac{1}{b-a} \left[\frac{b^2 - a^2}{2} \right] \\ &= \frac{1}{\cancel{b-a}} \left[\frac{(\cancel{b-a})(b+a)}{2} \right] \\ &= \frac{a+b}{2} \quad \square \end{aligned}$$

8. (3 pts) The fraction of small businesses that show an annual profit in a given year is a random variable whose probability density function is given by

$$f(y) = \begin{cases} ky^2(1-y), & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}.$$

What is the distribution of Y ? Handout 16, pg. 1

$$Y \sim \text{Beta}(3, 2)$$